ODE Definitions and Terminology

- An **IVP** is an **ODE** relating $\boldsymbol{y}(t)$ to $\boldsymbol{y}'(t), \boldsymbol{y}''(t), \ldots$ with an **IC** $\boldsymbol{y}(t_0) = \boldsymbol{y}_0$
  - **Order** is the highest derivative order.
  - A **Solution** satisfies ODE and IC for $a < t_0 < b$.
  - No explicit t dependence is **Autonomous**. An **Equilibrium** $f(y^*) = 0$ gives a constant solution $\boldsymbol{y}(t) = \boldsymbol{y}_*$.
- **Linear** means $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{b}(t)$, with coefficients independent of $\boldsymbol{y}$
  - **Constant Coefficient (CC)** means $\boldsymbol{A}$ and $\boldsymbol{b}$ are independent of t
  - $\boldsymbol{b}(t)$ is the **Forcing**. Zero forcing means **Homogeneous**
- First order scalar ODEs
  - **Separable** means $y' = g(t)h(y)$
  - **Standard Form** for 1st order linear is $y' + p(t)y = q(t)$.
- Modern numerical solvers give reliable numerical values for IVPs.

Linear Algebra: You should be able to . . .

- Set up and solve a **Linear System (LS)** $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ using $\boldsymbol{x}_c$ and $\boldsymbol{x}_p$.

$$\begin{aligned} 2x_1 - x_2 + x_3 &= 3 \\ x_1 + 3x_2 &= 7 \\ -x_2 + 2x_3 &= 4 \end{aligned} \quad \Longrightarrow \quad \underbrace{\begin{pmatrix} 2 & -1 & 1 \\ 1 & 3 & 0 \\ 0 & -1 & 2 \end{pmatrix}}_{\boldsymbol{A}} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_{\boldsymbol{x}} = \underbrace{\begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix}}_{\boldsymbol{b}}$$

- Compute eigenvalues $\lambda_1, \lambda_2, \ldots$ and eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots$ for $\boldsymbol{A} \in \mathbb{R}^{n \times n}$.

$$\boldsymbol{v}_i \neq \boldsymbol{0} \quad \& \quad \boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i \qquad \Longrightarrow \qquad \boldsymbol{A} \underbrace{\begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots \end{pmatrix}}_{\boldsymbol{V}} = \boldsymbol{V} \underbrace{\operatorname{diag}(\lambda_1, \lambda_2, \ldots)}_{\Lambda}$$

  - Be able to interpret complex $\alpha \pm \beta i$ eigenvalue pairs.
  - Remember what **Linear Independent (LI)** means.
- Use a computational tool for Linear Algebra.

Complementary and Particular Solutions $Ax = b$

$$\begin{aligned} x_1 + x_2 + x_3 &= 3 \\ x_2 - x_3 + x_4 &= 1 \\ x_1 + 2x_2 + x_4 &= 4 \\ 2x_1 + x_2 + 3x_3 - x_4 &= 5 \end{aligned} \implies \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 \\ 1 & 2 & 0 & 1 \\ 2 & 1 & 3 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 4 \\ 5 \end{pmatrix}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 3 \\ 0 & 1 & -1 & 1 & 1 \\ 1 & 2 & 0 & 1 & 4 \\ 2 & 1 & 3 & -1 & 5 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 0 & 2 & -1 & 2 \\ 0 & 1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \implies \boldsymbol{x} = \underbrace{\begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}}_{\boldsymbol{x}_p} + \underbrace{\begin{pmatrix} -2x_3 + x_4 \\ x_3 - x_4 \\ x_3 \\ x_4 \end{pmatrix}}_{\boldsymbol{x}_c}$$

$x_3, x_4 \in \mathbb{R}$ are the free variables.

Eigenvalues $A\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$ for $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$

- Eigenvalues satisfy $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$

  $$\det \begin{pmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} = (1-\lambda)^2 - 4 = (\lambda - 3)(\lambda + 1) = 0 \quad \Longrightarrow \quad \lambda_1 = 3,\ \lambda_2 = -1$$

- For $\lambda_1 = 3$, solve $(\boldsymbol{A} - 3\boldsymbol{I})\boldsymbol{x} = \boldsymbol{0}$

  $$\begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \boldsymbol{0} \implies x_1 = x_2 \implies \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- For eigenvectors $\boldsymbol{v}_1 = (1, 1)$ and $\boldsymbol{v}_2 = (1, -1)$

  $$A\boldsymbol{v}_1 = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\boldsymbol{v}_1 \qquad A\boldsymbol{v}_2 = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -\boldsymbol{v}_2$$

  - $\boxed{\lambda_1 = 3}$ so $\boldsymbol{A}$ preserves the direction of $\boldsymbol{v}_1$ and triples its length.
  - $\boxed{\lambda_2 = -1}$ so $\boldsymbol{A}$ reverses the direction of $\boldsymbol{v}_2$ while preserving its length.

$u' = au$ **vs** $\boldsymbol{y}' = A\boldsymbol{y}$

- The scalar equation $u' = au$ has solution $u(t) = ce^{at}$
- Try a solution of the form $\boldsymbol{y}_c(t) = e^{rt}\boldsymbol{v}$, with constant r and $\boldsymbol{v} \neq \boldsymbol{0}$
  - Substitute into $\boldsymbol{y}' = A\boldsymbol{y}$ and cancel e^{rt}

$$re^{rt}\boldsymbol{v} = e^{rt}A\boldsymbol{v} \quad \Longrightarrow \quad A\boldsymbol{v} = r\boldsymbol{v}$$

  - The form $\boldsymbol{y}_{c,i}(t) = e^{\lambda_i t}\boldsymbol{v}_i$ is a solution if λ_i and $\boldsymbol{v}_i$ are an eigenvalue–eigenvector pair
- The ODE $\boldsymbol{y}' = A\boldsymbol{y}$ is linear and homogeneous. If $A\boldsymbol{v}_1 = \lambda_1\boldsymbol{v}_1$ and $A\boldsymbol{v}_2 = \lambda_2\boldsymbol{v}_2$ a more general solution of $\boldsymbol{y}' = A\boldsymbol{y}$ is the linear combination

$$\boldsymbol{y}_c(t) = c_1e^{\lambda_1 t}\boldsymbol{v}_1 + c_2e^{\lambda_2 t}\boldsymbol{v}_2$$

- With n LI eigenvectors $A\boldsymbol{v}_i = \lambda_i\boldsymbol{v}_i$ all solutions are given by the **Superposition**

$$\boldsymbol{y}_c(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t}\boldsymbol{v}_i$$

First Order Homogeneous Linear System $y' = Ay$

- For constant $\boldsymbol{A} \in \mathbb{R}^{n \times n}$, compute eigenpairs

$$\boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i \qquad \boldsymbol{v}_i \neq \boldsymbol{0}$$

- Each eigenpair gives a solution

$$\boldsymbol{y}_{c,i}(t) = e^{\lambda_i t}\boldsymbol{v}_i \qquad \boldsymbol{y}'_{c,i}(t) = \lambda_i e^{\lambda_i t}\boldsymbol{v}_i = \boldsymbol{A}\boldsymbol{y}_{c,i}(t)$$

- If $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ are linearly independent, the **General Solution (GS)** is

$$\boxed{\boldsymbol{y}_c(t) = c_1 e^{\lambda_1 t}\boldsymbol{v}_1 + \cdots + c_n e^{\lambda_n t}\boldsymbol{v}_n}$$

- The IC $\boldsymbol{y}(0) = \boldsymbol{y}_0$ determines $\boldsymbol{c} = (c_1, \ldots, c_n)$ through the LS

$$\boldsymbol{V}\boldsymbol{c} = \boldsymbol{y}_0 \qquad \text{with} \quad \boldsymbol{V} = (\boldsymbol{v}_1 \ \cdots \ \boldsymbol{v}_n)$$